

Complexity Explorables

Lotka Martini: the lotka-volterra model

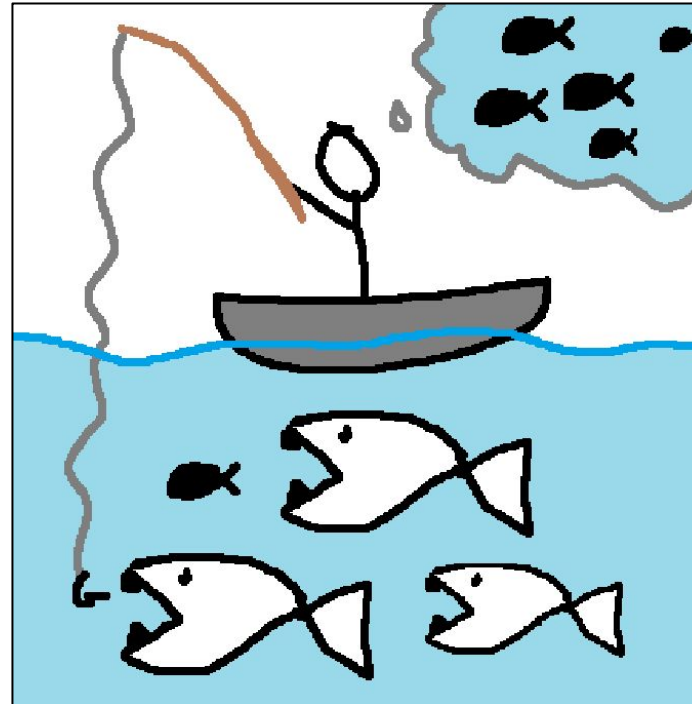
Outline

- Historic Motivation
- The Lotka Volterra Model
 - ◆ Basic assumptions
 - ◆ Differential equations
 - ◆ Behavior and Fixpoints
- How does the Simulation work
- Examples
- Discussion

Historic Motivation

• **1910:** proposed by Alfred J. Lotka (for plants and herbivores)

• **1926:** same equations published by Vito Volterra (fishing)



Symbolic image of Umberto D'Ancona

Basic Assumptions

Prey(X):

- ❖ *Reproduction*(α_R): $X + \emptyset \rightarrow 2X$
- ❖ *Natural death*(α_D): $X \rightarrow \emptyset$
- ❖ *Predation*(β): $X + 2Y \rightarrow \emptyset$

Predators(Y):

- ❖ *Reproduction*(β): $2Y + X \rightarrow 3Y$
- ❖ *Death*(γ): $Y \rightarrow \emptyset$

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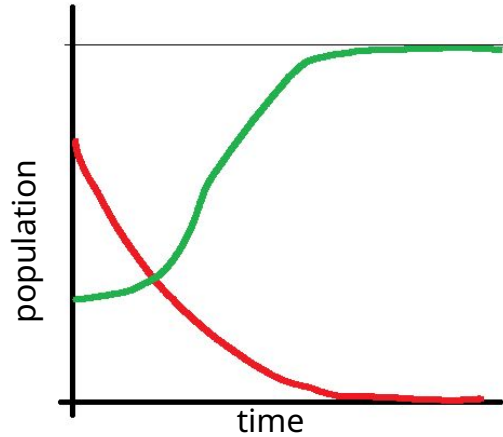
Lotka-Volterra equations:

$$\dot{X} = \alpha X - \beta X Y$$

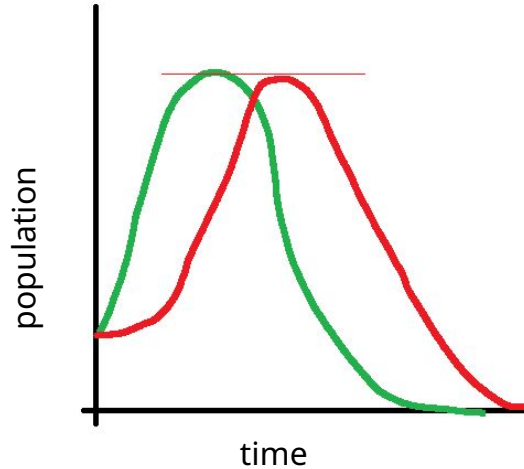
$$\dot{Y} = -\gamma Y + \beta X Y$$

$$\alpha = \alpha_R - \alpha_D > 1$$

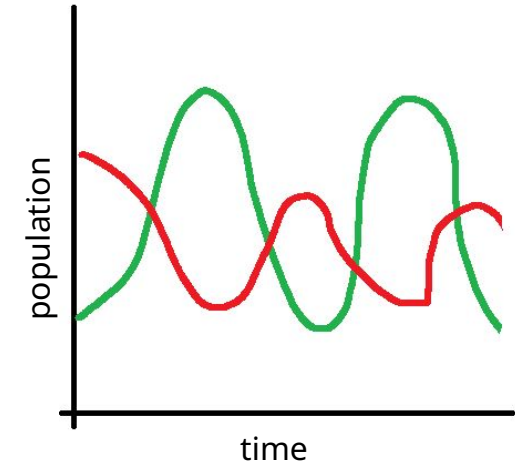
Behavior and Fixpoints



- ⇒ Predators die out
- ⇒ no natural cause of death for prey ($\alpha > 1$)
- ⇒ Prey multiplies until full capacity



- ⇒ Number of predators exceeds number of Prey
- ⇒ Prey dies off
- ⇒ Predators can not multiply and die off as well



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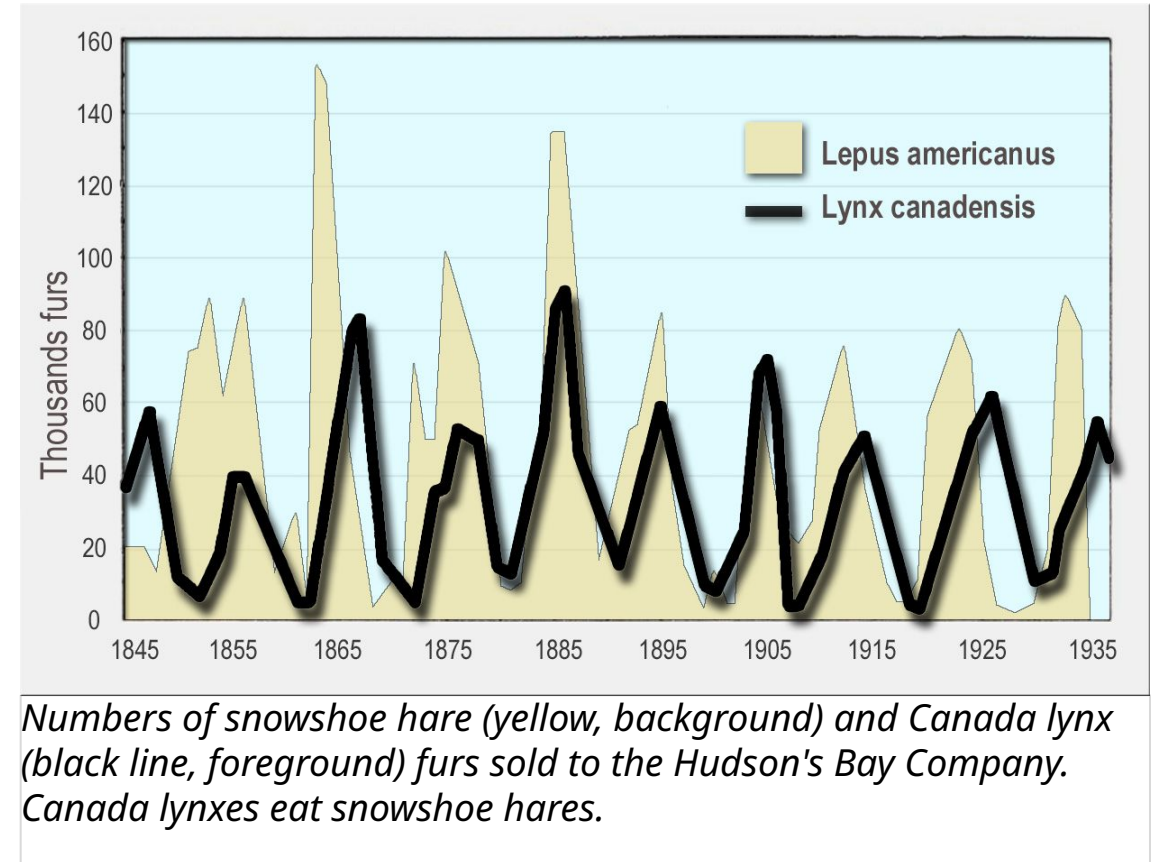
Periodic behavior

- oscillating numbers
- self-regulating:
 - equilibrium densities:

$$X^* = \gamma / \beta$$

$$Y^* = \alpha / \beta$$

⇒ change of X/Y depends on the growth/decay parameters of Y/X



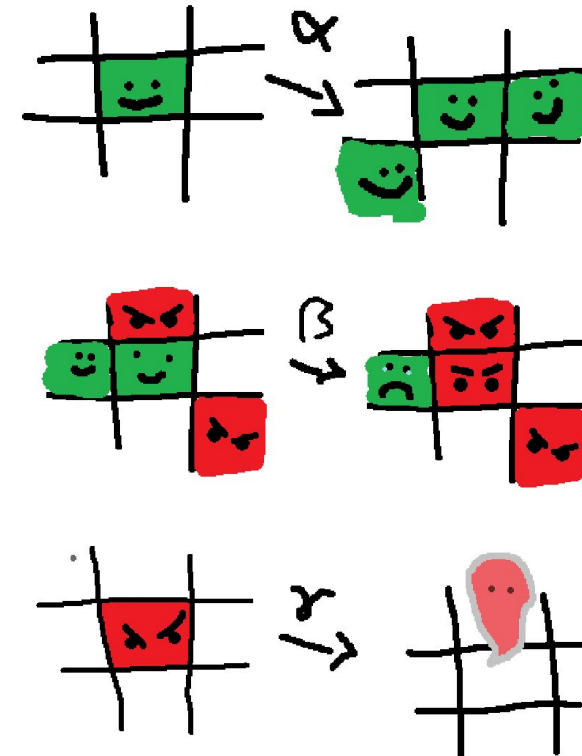
How does the Simulation work

Prey(X):

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- ❖ **Natural death(α_D):** $X \rightarrow \emptyset$
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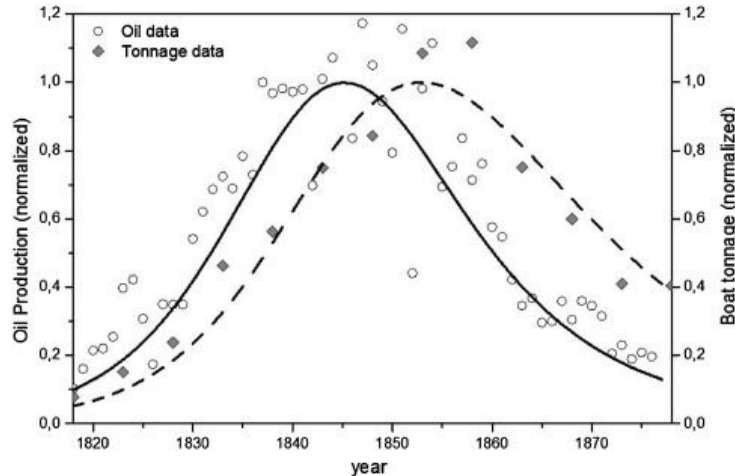
Predators(Y):

- ❖ **Reproduction(β):** $2Y + X \rightarrow 3Y$
- ❖ **Death(γ):** $Y \rightarrow \emptyset$



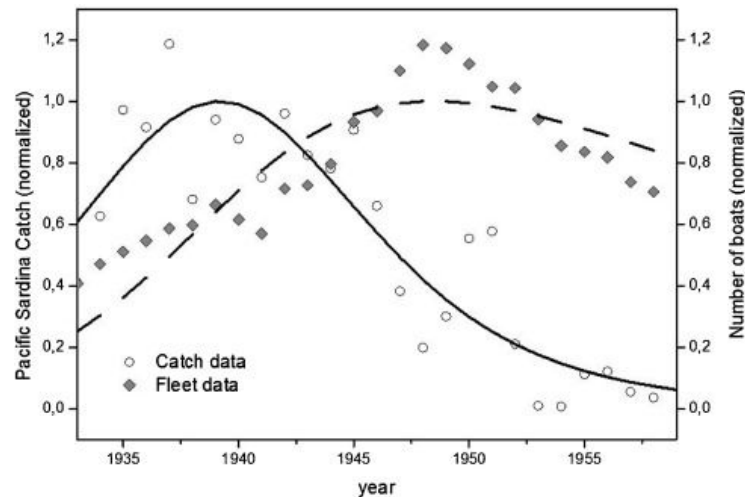
"Dynamic patterns of overexploitation in fisheries"

by Ilaria Perissi, Ugo Bardi, Toufic El Asmar, Alessandro Lavacchi



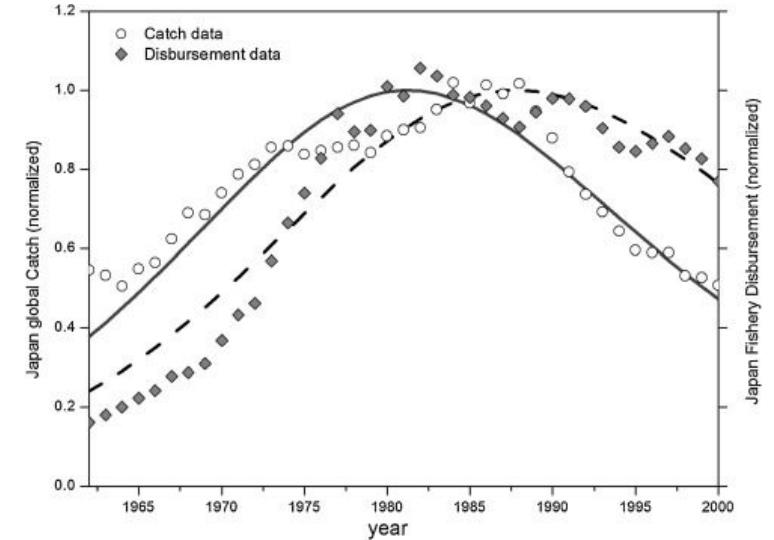
Lotka-Volterra modeling of **American Whaling** on the 19th century.

Sperm Oil (production-prey) and the Tonnage capacity of fishing boats (capital-predator) from 1818 to 1878. Normalizing factors: oil $1.16 \cdot 10^5$ gallons, Boat tonnage $9.72 \cdot 10^4$.



Lotka Volterra modeling of **Californian Sardines Fishery**.

Catch (production-prey) and the number of fishing boats (capital-predator) from 1933 to 1959. Normalizing factors: catch $6.12 \cdot 10^5$ Tons, the number of boats $5.21 \cdot 10^3$.



Lotka Volterra modeling of **Japanese Fishery Industry**.

Total Catch (production-prey) and the Disbursement of Fishery (capital-predator) from 1962 to 2000. Normalizing factors: catch $1.26 \cdot 10^7$ Tons, disbursement $1.35 \cdot 10^8$ Yen.

“Dynamics of collective attention”

by Philipp Lorenz-Spreen

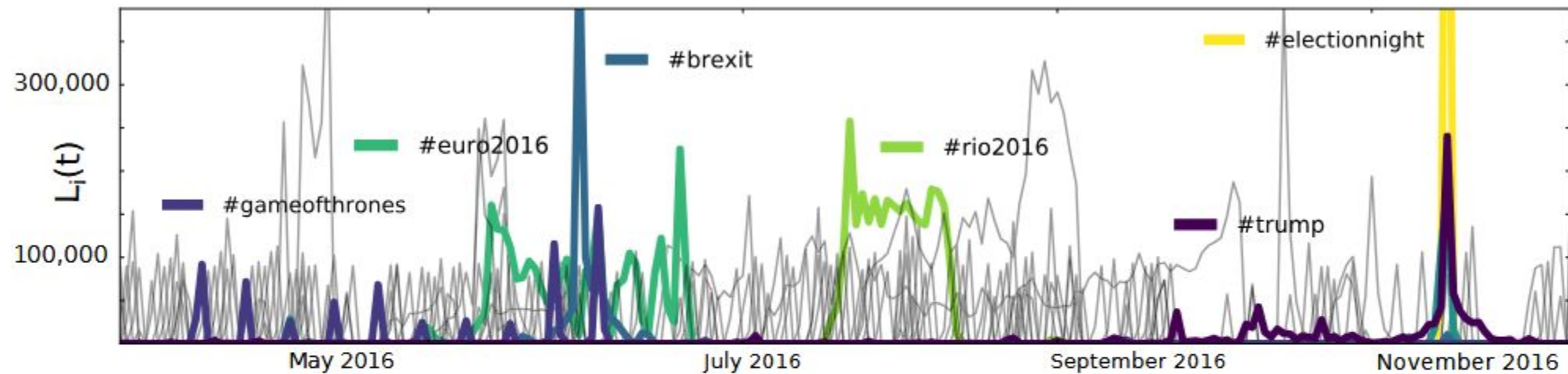


FIGURE 7.1: **Popular hashtag usage over one year on Twitter:** a Daily usage $L_i(t)$ of a random sample from the top 50 hashtags i on Twitter in 2016, with exemplary highlighted major events.

Discussion

Thank you for your attention.